UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Classes 003–004 · Total derivative, chain rule and implicit functions · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S003·7 · S004·9	Deck 003 / 004, slide number (rendered .pptx)
T003a 00:06:27	Transcript 003 part I: “Total derivative, chain rule and implicit functions”
T003b 00:01:10	Transcript 003 part II: “Problems on total derivatives, chain rule”
T004a 00:13:09	Transcript 004 part I: “Implicit functions and problems”
T004b 00:02:31	Transcript 004 part II: “Problems on composite and implicit functions”
CW U1 P14 · ESA Jan-24 s3 1b	Unit 1 class-work problem 14 · past-paper question

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A Formula sheet

1 Composite functions and the one-variable chain rule

E1.1 · Chain rule, one variable

ESA ★

$$\frac{d}{dx}\,f\big(g(h(x))\big) = f'\big(g(h(x))\big)\cdot g'\big(h(x)\big)\cdot h'(x)$$

GIVES Derivative of a composition: outer to inner, multiplied.

APPLIES Each function differentiable.

NOTE Example: $\dfrac{d}{dx}e^{\sin x^2} = e^{\sin x^2}\cos x^2\cdot 2x$.

S003·4 · T003a 00:00:33

2 The total derivative rule

E2.1 · Total derivative rule

ESA ★★

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

GIVES Rate of change of z with respect to the single variable t.

APPLIES $z = f(x,y)$, $x = x(t)$, $y = y(t)$.

NOTE Ordinary d on z, x, y with respect to t; partial ∂ only on z with respect to x, y.

S003·5 · T003a 00:05:54 · T003b 00:01:10

E2.2 · Total derivative, three variables

ESA ★★

$$\frac{df}{dt} = f_x\frac{dx}{dt} + f_y\frac{dy}{dt} + f_z\frac{dz}{dt}$$

GIVES One term per intermediate variable.

APPLIES $f = f(x,y,z)$, each of x, y, z a function of t.

NOTE Used in CW U1 P11 and in the triangle problem with x, y, θ.

ADDED FOR CLARITY S003·9 · CW U1 P11

3 The total differential: approximation and sensitivity

E3.1 · Total differential

ESA ★★

$$dz = f_x(x,y)\,dx + f_y(x,y)\,dy$$

GIVES Approximate change in z for small changes dx, dy.

APPLIES $z = f(x,y)$ differentiable.

NOTE n variables: $dz = f_{x_1}dx_1 + f_{x_2}dx_2 + \cdots + f_{x_n}dx_n$.

S004·3 · S004·4 · S003·8 · T003a 00:13:56

E3.2 · Approximating a nearby value

ESA ★★

$$f(x_0+dx,\,y_0+dy) \approx f(x_0,y_0) + f_x(x_0,y_0)\,dx + f_y(x_0,y_0)\,dy$$

GIVES Estimated function value near a known point.

APPLIES dx, dy small.

NOTE Slide 003·8: $f(2.1,-3.03) \approx 6 + 1.3(0.1) + (-0.6)(-0.03) = 6.148$.

S003·8 · T003a 00:13:09

4 Rates of change problems

E4.1 · Constant quantity

ESA ★★

$$A = \text{constant} \implies \frac{dA}{dt} = 0$$

GIVES The equation that fixes the unknown rate.

APPLIES "Remains constant" in the question.

S003·9 · T003a 00:16:32

5 The chain rule for composite functions

E5.1 · Chain rule, two intermediate variables

ESA ★★★

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u}, \qquad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

GIVES Partial derivatives of a composite function.

APPLIES $z = f(x,y)$, $x = g(u,v)$, $y = h(u,v)$.

NOTE All derivatives are partial, because everything depends on two variables.

S003·13 · S003·14 · T003a 00:21:07

E5.2 · Chain rule, three intermediate variables

ESA ★★★

$$\frac{\partial v}{\partial x} = \frac{\partial v}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial v}{\partial s}\frac{\partial s}{\partial x} + \frac{\partial v}{\partial t}\frac{\partial t}{\partial x}$$

GIVES One term per intermediate variable.

APPLIES $v = f(r, s, t)$, each of r, s, t a function of x, y, z.

NOTE A term is 0 when that intermediate variable does not contain x.

S003·15 · T003a 00:23:10

E5.3 · Function of one intermediate variable

ESA ★★

$$u = f(r),\ r = \sqrt{x^2 + y^2 + z^2} \implies \frac{\partial u}{\partial x} = f'(r)\,\frac{\partial r}{\partial x} = f'(r)\,\frac{x}{r}$$

GIVES $\partial u/\partial x$ in terms of $f'(r)$.

APPLIES u depends on x, y, z only through r.

S004·12 · T004b 00:05:07

E5.4 · Polar coordinates: derivatives you reuse

ESA ★★

$$\begin{aligned} &x = r\cos\theta,\ y = r\sin\theta: && x_\theta = -r\sin\theta, \quad y_\theta = r\cos\theta \\ &r = \sqrt{x^2+y^2},\ \theta = \tan^{-1}\tfrac{y}{x}: && r_x = \frac{x}{r},\ r_y = \frac{y}{r},\ \theta_x = -\frac{y}{r^2},\ \theta_y = \frac{x}{r^2} \end{aligned}$$

GIVES The inner derivatives for polar problems.

APPLIES $r = \sqrt{x^2+y^2}$, $\theta = \tan^{-1}(y/x)$.

NOTE So $u_x = f_r\frac{x}{r} - f_\theta\frac{y}{r^2}$ and $u_y = f_r\frac{y}{r} + f_\theta\frac{x}{r^2}$ for $u = f(r,\theta)$.

S004·13 · S004·15 · S004·17

E5.5 · Polar identity

ESA ★★

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = \left(\frac{\partial u}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial u}{\partial \theta}\right)^2$$

GIVES The squared gradient in polar coordinates.

APPLIES $u = u(x, y)$, $x = r\cos\theta$, $y = r\sin\theta$.

NOTE Proved in W5.16.

CW U1 P15

6 Implicit functions of two variables

E6.1 · Implicit differentiation, two variables

ESA ★★

$$f(x,y)=c \implies \frac{dy}{dx}=-\frac{f_x}{f_y}, \qquad f_y\neq 0$$

GIVES Slope of the level curve at a point.

APPLIES $f_y \neq 0$ at the point.

NOTE From $df = f_x\,dx + f_y\,dy = 0$. Slide 004·4 prints the fraction upside down.

S004·5 · S004·6 · T004a 00:06:36 · T003b 00:10:33

E6.2 · Tangent line at a point

ESA ★★

$$y-y_0=m\,(x-x_0), \qquad m=\left.\frac{dy}{dx}\right|_{(x_0,y_0)}$$

GIVES Equation of the tangent to the curve at (x_0, y_0).

APPLIES (x_0, y_0) on the curve.

NOTE Through the origin: $y = mx$.

S004·7 · T004a 00:14:25

E6.3 · Derivative of a constant power and a variable power

ESA ★★

$$\frac{d}{dx}a^x=a^x\ln a, \qquad \frac{d}{dx}x^n=n\,x^{n-1}$$

GIVES The two rules needed for $(\cos x)^y$ and $(\sin y)^x$.

APPLIES In a partial derivative, the other variable is a constant.

NOTE $\ln$ needs a positive argument: $\cos x > 0$, $\sin y > 0$.

S004·8 · T004a 00:16:43

7 Implicit functions of three variables

E7.1 · Implicit differentiation, three variables

ESA ★★

$$f(x,y,z)=0: \quad \frac{\partial z}{\partial x}=-\frac{f_x}{f_z}, \quad \frac{\partial z}{\partial y}=-\frac{f_y}{f_z}\ (f_z\neq 0); \qquad \frac{\partial y}{\partial x}=-\frac{f_x}{f_y}\ (f_y\neq 0)$$

GIVES Partial derivatives of the implicitly defined variable.

APPLIES Hold y constant for z_x; hold z constant for y_x.

NOTE Extends to any number of variables.

S004·6 · S003·26 · T004a 00:10:23 · T003b 00:09:29

B Symbol glossary

Symbol	Name	Meaning	Appears in
$z = f(x, y)$	Function of two variables	The quantity being differentiated	E2.1, E3.1, E5.1
x, y	Intermediate variables	Depend on t (total derivative) or on u, v (chain rule)	E2.1, E5.1
t	Parameter / time	Single final variable in the total derivative; also a new variable in some chain-rule problems	E2.1, E2.2
u, v / s, t	New independent variables	Final variables of a composite function	E5.1
r, s, t	Named arguments	$r = 2x - 3y$, ... in slide 003·15 (not a radius there)	E5.2
r, θ	Polar coordinates	$x = r\cos\theta$, $y = r\sin\theta$	E5.4, E5.5
$\dfrac{dz}{dt}$	Total derivative	Ordinary derivative when z ends up depending on t alone	E2.1
$\dfrac{\partial z}{\partial u}$	Partial derivative of a composite	z depends on u and v	E5.1
$dx,\ dy,\ dz$	Differentials	Small (signed) changes; $dz = f_x\,dx + f_y\,dy$	E3.1, E3.2
f_x, f_y, f_z	Partial derivatives of f	Kept as symbols when f is unknown	E5.1, E6.1, E7.1
$f(x, y) = c$	Level curve / implicit relation	Defines y as a function of x	E6.1
$f(x, y, z) = 0$	Level surface	Defines z as a function of x, y	E7.1
$f'(r)$	Ordinary derivative of $f(r)$	u depends on x, y, z only through r	E5.3
m	Slope	dy/dx at the point; tangent $y - y_0 = m(x - x_0)$	E6.2
A, V, h, b, θ	Geometric quantities	Area, volume, height, base, included angle in rate problems	E4.1

Remark. Watch the letters that change meaning between problems: t is the parameter in the total derivative but one of the new variables in the chain-rule problems (s, t); r, s, t are intermediate variables in slide 003·15 but r is a radius elsewhere; f is the outer function in composite problems and the implicit relation $f(x, y) = 0$ in class 004.

C Ordinary d or partial ∂?

Situation	Write	Why
$z = f(x,y)$, $x = x(t)$, $y = y(t)$	$\dfrac{dz}{dt}$ (total)	z ends up depending on one variable t
$z = f(x,y)$, $x = x(u,v)$, $y = y(u,v)$	$\dfrac{\partial z}{\partial u}$, $\dfrac{\partial z}{\partial v}$	z ends up depending on two variables
$f(x,y) = 0$	$\dfrac{dy}{dx} = -\dfrac{f_x}{f_y}$	y is a function of x alone
$f(x,y,z) = 0$	$\dfrac{\partial z}{\partial x} = -\dfrac{f_x}{f_z}$	z is a function of x and y
$z = f(x,y)$	$dz = f_x\,dx + f_y\,dy$	total differential: a small change